

Checking Ritter's dry dam break

Substitution into the one-dimensional shallow water equations

A subscript on a field is a partial derivative. The depth is h , the depth-averaged velocity is u , and g is gravity. The bed is flat, so $z = 0$ and the free surface is the depth. This note takes Ritter's dry dam break from the companion note and substitutes it into the shallow water equations.

1. The equations

On a flat frictionless bed the system is

$$\begin{aligned} h_t + (hu)_x &= 0, \\ u_t + uu_x + gh_x &= 0. \end{aligned} \tag{1}$$

Where $h > 0$ this is equivalent to the conservation form

$$\begin{aligned} h_t + (hu)_x &= 0, \\ (hu)_t + \left(hu^2 + \frac{gh^2}{2} \right)_x &= 0. \end{aligned} \tag{2}$$

The second form is the one that stays meaningful where the depth reaches zero.

2. The candidate

Let $c_L = \sqrt{gh_L}$ with $h_L > 0$, and write $\xi = x/t$ for $t > 0$. Ritter's solution is

$$h(x, t) = \begin{cases} h_L & \text{if } x < -c_L t \\ \frac{1}{9g}(2c_L - \xi)^2 & \text{if } -c_L t \leq x \leq 2c_L t \\ 0 & \text{if } x > 2c_L t \end{cases} \tag{3}$$

$$u(x, t) = \begin{cases} 0 & \text{if } x < -c_L t \\ \frac{2}{3}(c_L + \xi) & \text{if } -c_L t \leq x \leq 2c_L t \\ 0 & \text{if } x > 2c_L t. \end{cases} \tag{4}$$

The three regions are checked separately. On the ray $x = -c_L t$ and on the dry front $x = 2c_L t$ the piecewise formulas meet, so those two lines are checked afterwards.

The fields depend on x and t only through ξ . Against $x/(c_L t)$ the depth is measured in units of h_L and the velocity in units of c_L . Still water stands to the left of -1 , the rarefaction occupies $-1 \leq x/(c_L t) \leq 2$, and the bed is dry to the right of the front. The velocity is drawn on the wet bed only. It is zero in the still water and equals $2c_L$ at the front.

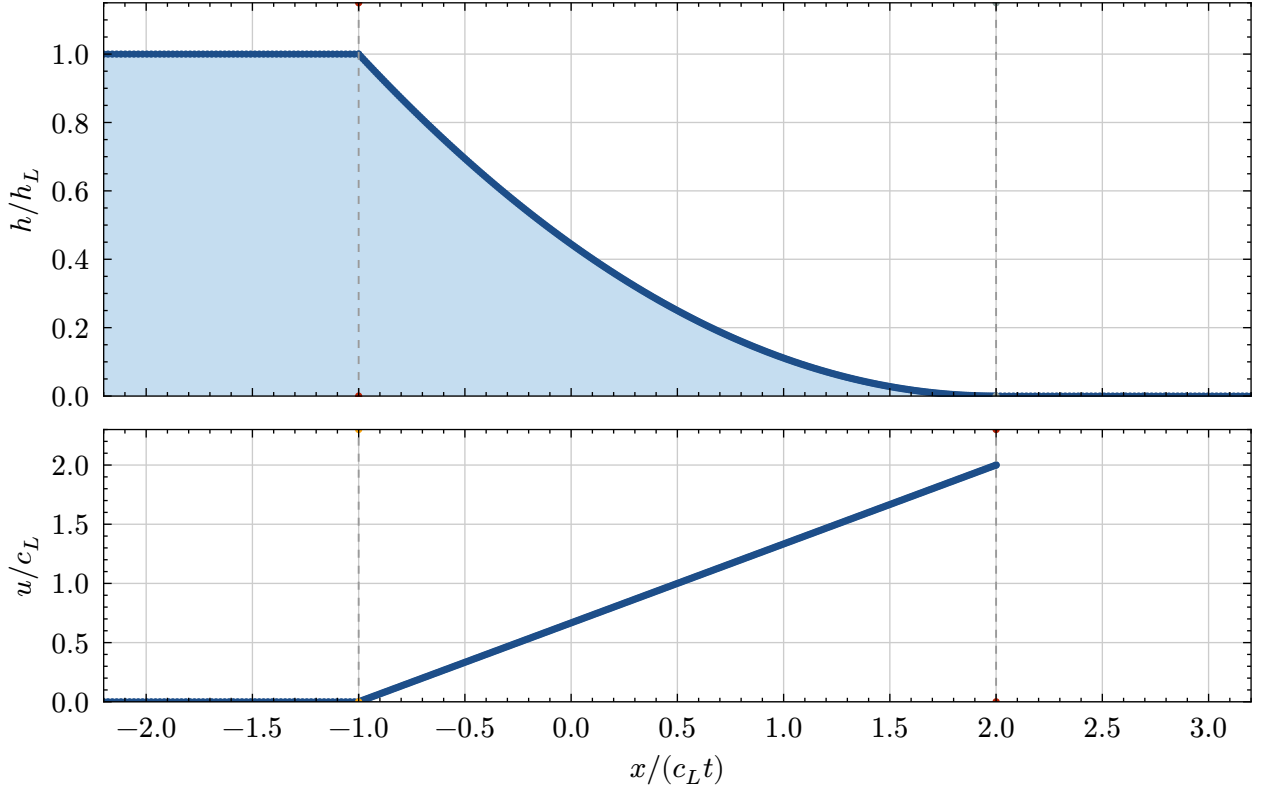


Figure 1: Ritter's solution at a fixed $t > 0$. The dashed lines are $x = -c_L t$ and the dry front $x = 2c_L t$.

3. Still water

For $x < -c_L t$ the fields are the constants $h = h_L$ and $u = 0$. Every derivative is zero, so both equations reduce to $0 = 0$.

4. The rarefaction

Inside $-c_L t < x < 2c_L t$, set $\varphi = 2c_L - \xi$. The formulas become

$$u = \frac{2}{3}(c_L + \xi), \quad h = \frac{\varphi^2}{9g}, \quad c = \sqrt{gh} = \frac{\varphi}{3}. \quad (5)$$

In particular $u + 2c = 2c_L$ throughout the fan. Since $\xi_t = -\xi/t$ and $\xi_x = 1/t$,

$$u_t = -\frac{2\xi}{3t}, \quad u_x = \frac{2}{3t}, \quad h_t = \frac{2\varphi\xi}{9gt}, \quad h_x = -\frac{2\varphi}{9gt}. \quad (6)$$

Mass is the expansion $(hu)_x = h_x u + h u_x$, so

$$h_t + (hu)_x = \frac{2\varphi}{9gt}(\xi - u) + \frac{\varphi^2}{9g} \frac{2}{3t} = \frac{2\varphi}{9gt} \left(\xi - u + \frac{\varphi}{3} \right). \quad (7)$$

The parenthesis is

$$\xi - \frac{2}{3}(c_L + \xi) + \frac{2c_L - \xi}{3} = \xi - \frac{2}{3}\xi - \frac{1}{3}\xi = 0. \quad (8)$$

Hence $h_t + (hu)_x = 0$.

Momentum is

$$u_t + uu_x + gh_x = -\frac{2\xi}{3t} + \frac{4}{9t}(c_L + \xi) - \frac{2\varphi}{9t} = \frac{1}{9t}(-6\xi + 4c_L + 4\xi - 2\varphi). \quad (9)$$

With $\varphi = 2c_L - \xi$ the numerator is $-2\xi + 4c_L - 2(2c_L - \xi) = 0$. Hence $u_t + uu_x + gh_x = 0$.

5. Dry bed

For $x > 2c_L t$ one has $h = 0$. The conserved quantities are h and hu , and both are zero, so both fluxes hu and $hu^2 + gh^2/2$ are zero. Their derivatives vanish and the conservation form holds. The value assigned to u on dry ground does not enter those fluxes.

6. The two edges

At $x = -c_L t$, one has $\xi = -c_L$, so the rarefaction formulas give $u = 0$ and $h = c_L^2/g = h_L$. Depth and velocity match the still water. The derivatives do not: just inside the fan, $u_x = 2/(3t)$ and $h_x = -2c_L/(3gt)$. The fields are continuous across this ray and the equations hold on either side, so the ray carries no jump.

At $x = 2c_L t$, one has $\xi = 2c_L$, so $h = 0$ and $u = 2c_L$. The discharge is $hu = 0$, which matches the dry bed. The front advances at speed $2c_L$, the same speed as the fluid at the front. It is the wet-dry boundary of the rarefaction, and the conserved variables (h, hu) stay continuous across it.

Inside each open region the shallow water equations hold, the still water joins the rarefaction continuously, and the dry front carries no jump in h or hu . Ritter's formula is a solution.